1. Data Preprocessing
Write a Python program to preprocess the Titanic dataset (titanic.csv)

(a) Show the dataset's basic details: its size, column names, data types, and summary statistics.

(b)Display the number of missing values in each column.
(b) Fill missing Age values with the median and Embarked values with the mode.
(c) Identify the target column and drop and save to a new column.
(c) Identify the categorical features and convert to numeric form using label encoding.
(d) Identify the outliers using boxplot. Apply Min-Max normalization to remove outliers.

(e) Split the data into training and test sets and print the shapes of both.
Print the first five rows of the dataset after each step.

2. Principal Component Analysis (PCA)
Write a Python program to apply PCA on the Iris dataset to reduce it from 4 features to 2:
(a) Standardize and plot a scatter plot before applying PCA.
(b) Print the explained variance ratio of each principal component and the total variance retained.
(c) Print the first five rows of the reduced data.
(d) Plot the reduced data as a 2D scatter plot with a different colour for each species.

3. Write a program to perform logistic regression on the Wine dataset.

4. Support Vector Machine (SVM)
Write a Python program to classify tumours as malignant or benign using SVM:
(a) Scale the features using StandardScaler.
(b) Train two SVM models using linear and RBF kernels and print the accuracy of both.
(c) Display the confusion matrix and print accuracy,precision, recall, F1-score

(d) Print the classification report.
Dataset: Breast Cancer Wisconsin (sklearn.datasets.load_breast_cancer)

5. K-Nearest Neighbours (KNN)
Write a Python program to implement KNN classification on the Iris dataset
(a) Scale the features using StandardScaler.
(b) Classify the test set and print the correct and wrong predictions.
(c) Vary k from 1 to 15, plot accuracy against k, and report the best value of k.

6. Decision Tree
Write a classification program for implementing decision tree using the

playtennis dataset
(a) Encode the categorical attributes and build the decision tree.
(b) Print the tree in text form using export_text and display it graphically using plot_tree.
(c) Print the accuracy of the tree on the training data.
(d) Classify the new sample: Outlook = Sunny, Temperature = Cool, Humidity = High, Wind = Strong.
Dataset: playtennis.csv (14 rows)

7. K-Means Clustering
Write a Python program to implement K-Means clustering using the Mall Customers dataset

(a) Use the Elbow method (k = 1 to 10) to find the optimal number of clusters and plot the elbow curve.
(b) Apply K-Means with the optimal k and plot the clusters with their centroids marked.

8. Artificial Neural Network (ANN)
Write a Python program to build a multilayer perceptron using MLPClassifier to predict diabetes:
(a) Scale the features using StandardScaler.
(b) Build the network with two hidden layers (16 and 8 neurons) using ReLU activation and train for up to 500 iterations.
(c) Plot the training loss curve using loss_curve_.
(d) Print the test accuracy and confusion matrix.
Dataset: diabetes.csv (Pima Indians Diabetes, 768 rows)

9. Convolutional Neural Network (CNN)
Write a Python program to build a CNN using Keras to classify handwritten digits:
(a) Reshape the images to 8×8×1 and normalize the pixel values.
(b) Build the model with one Conv2D layer (16 filters, 3×3), one MaxPooling layer, a Flatten layer, a Dense hidden layer and a softmax output layer.
(c) Train for 20 epochs and plot the accuracy curve.
(d) Print the test accuracy and the predicted versus actual labels for five test images.
Dataset: Digits (sklearn.datasets.load_digits, 1,797 images of 8×8 pixels)